

THE UNCERTAINTY OF THE PHONY

Philosophers in the wrong places—Uncertainty about (mostly) lunch—What I don't care about—Education and intelligence

This final chapter of Part Three focuses on a major ramification of the ludic fallacy: how those whose job it is to make us aware of uncertainty fail us and divert us into bogus certainties through the back door.

LUDIC FALLACY REDUX

I have explained the ludic fallacy with the casino story, and have insisted that the sterilized randomness of games does not resemble randomness in real life. Look again at [Figure 7](#) in [Chapter 15](#). The dice average out so quickly that I can say with certainty that the casino will beat me in the very near long run at, say, roulette, as the noise will cancel out, though not the skills (here, the casino's advantage). The more you extend the period (or reduce the size of the bets) the more randomness, by virtue of averaging, drops out of these gambling constructs.

The ludic fallacy is present in the following chance setups: random walk, dice throwing, coin flipping, the infamous digital “heads or tails” expressed as 0 or 1, Brownian motion (which corresponds to the movement of pollen particles in water), and similar examples. These setups generate a quality of randomness that does not even qualify as randomness—*protorandomness* would be a more appropriate designation. At their core, all theories built around the ludic fallacy ignore a layer of uncertainty. Worse, their proponents do not know it!

One severe application of such focus on small, as opposed to large, uncertainty concerns the hackneyed *greater uncertainty principle*.

Find the Phony

The greater uncertainty principle states that in quantum physics, one cannot measure certain pairs of values (with arbitrary precision), such as the position and momentum of particles. You will hit a lower bound of measurement: what you gain in the precision of one, you lose in the other. So there is an